Mix Scenarios Research and Documentation

Scenario #1 Portland Cement

Scenario #2 Flowable Fill

Flowable fill concrete is a self-compacting and fast settling concrete use for backfilling voids, trenches, and structures. It eliminates the need for compaction equipment and reduces labor. The concrete flows into tight spaces and sets quickly ("How Flowable Fill Works"). Flowable fill is a mixture of cement, fly ash, fine sand, water, and air ("2020 city of Lincoln"). The approximate compressive strength should range from 85 to 175 psi. No less than 95% of sand should pass through the No.4 sieve which is under the category of fine aggregate. No more than 5% should pass through the No.200 sieve which is the finest sieve.

Scenario #3 L-3500

L-3500 is a class of concrete that is a specific type of Portland Cement Concrete (PCC). It's general use is for pavements, sidewalks, and structures. This class of concrete is usually used in areas of high compressive strength being able to withstand significant loads (City of Lincoln 2020 Standard Specifications Chapter 4 – Portland Cement Concrete (PCC) Pavement). The minimum PSI strength for this mixture is 3500 psi.

Scenerio #4 LB-2750

LB-2750 is a class of concrete that is used for new construction of residential streets (City of Lincoln 2020 Standard Specifications Chapter 5 – Portland Cement Concrete (PCC) Base Construction). This is generally a pavement base with the minimum PSI being 2750.

Works Cited

2020 city of Lincoln Standard Specifications 300. (n.d.-a). https://www.lincoln.ne.gov/files/sharedassets/public/v/2/ltu/transportation/standards/standard-specifications/2020/chapter3.pdf

2020 city of Lincoln Standard Specifications 400. (n.d.-b). https://www.lincoln.ne.gov/files/sharedassets/public/v/1/ltu/transportation/standards/standard-specifications/2020/chapter4.pdf

2020 city of Lincoln Standard Specifications 500. (n.d.-c). https://www.lincoln.ne.gov/files/sharedassets/public/v/1/ltu/transportation/standards/standard-specifications/2020/chapter5.pdf

Flashfill Services. (2025, August 27). *How flowable fill works: Complete contractor guide 2025*. https://www.flashfillservices.com/post/how-flowable-fill-works